Adaptive Reading Budgets for Joint Forecast Audits with Cumulative Data

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Abstract

Background. Model audits share verification readings across predictors. We ask when choosing later readings after earlier answers reduces the number needed for precise comparisons.

Methods. We combine sharp comparison intervals, adaptive query planning and exact randomized-batch certificates, with replay on cumulative meters.

Results. Under compact convex uncertainty, for one comparison, if a deterministic policy guarantees precision with two exact scalar readings, some randomized fixed batch succeeds with worst-case probability at least $1/2$. This lower bound disappears with two comparisons. For every $m \geq 3$, we construct instances where adaptation always succeeds but the best randomized batch succeeds with worst-case probability exactly $1/m$. All coordinates are queryable, the budget remains two, and each comparison is separately certifiable by a fixed batch. The separation also holds for all pairwise comparisons among three models. On 225 of 1,098 meter blocks across six systems, two adaptive readings guarantee joint half-width precision where three fixed readings are necessary. In 191 cases, every scalar comparison separately needs at most two fixed readings, isolating a shared-budget obstruction. We certify optimal randomized-batch precision on all 1,098 blocks. Optional-reading reductions are 6.40% and 7.67% across the panels. Absolute-precision and capacity analyses identify when simpler fixed plans suffice. Unrestricted bulk retrieval is faster on the tested public interface.

Conclusions. Sharing a budget can make feedback necessary even when separate comparisons need none. The meter study identifies when feedback saves readings and when simpler protocols suffice.

Keywords: artificial intelligence; machine learning; adaptive information; minimax risk; optimal recovery; robust optimization.

1 Introduction

A model audit may need more verification data before it can report a precise comparison. The auditor can retrieve a complete record, choose a batch of readings, or choose later readings after seeing earlier answers. We study this last option, adaptive querying. Our output is a sharp interval:

the smallest interval containing the loss difference for every record compatible with the observations. Its width measures how much the reported comparison can still change. We ask when feedback is needed to meet a stated width requirement with the same reading budget. This is an acquisition problem for fixed predictors. The auditor specifies which loss contrasts must be precise, what information is already available, and which measurements may be requested. We first compute the smallest sufficient fixed reading budget, then test whether two adaptive readings can reduce it. The algorithm returns a reading plan and checks its intervals. Identifying a better model is a separate condition: a comparison interval must also exclude zero.

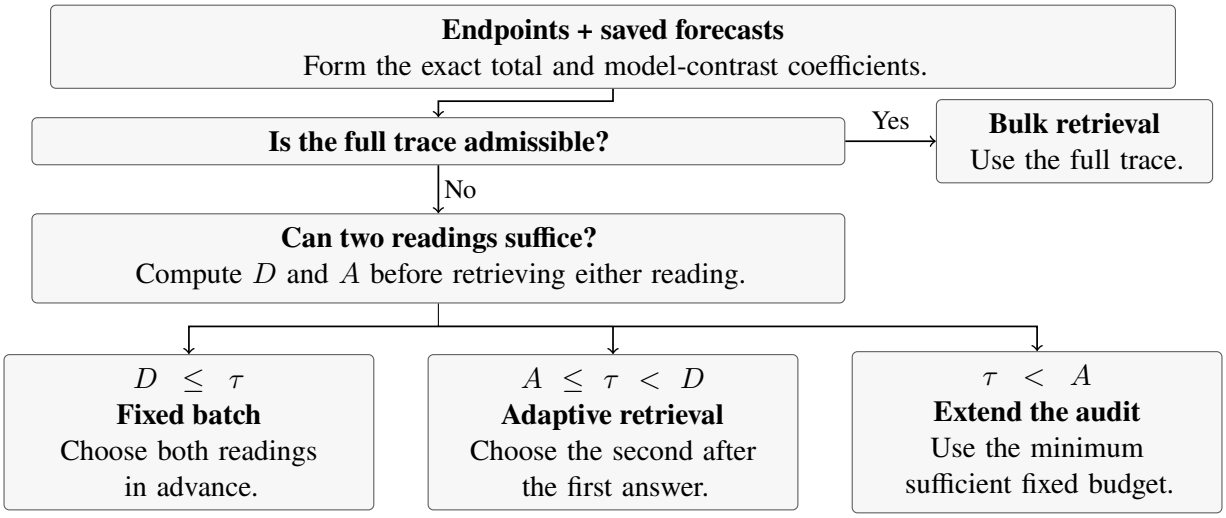


Figure 1: Choosing an audit protocol. For the exact-total, noiseless prefix model, let $J_Q = \max_r W_{r,Q}/W_{r,0}$ over contrasts with positive initial width. Let $D = \min_Q \sup_y J_Q(y)$ and $A = \min_{P \text{ det. adaptive}} \sup_y J_{Q_P(y)}(y)$ under the same two-reading budget. We fix the target τ before retrieval; the main application uses $\tau = 1/2$. Since $A \leq D$, the three restricted-access branches exhaust the possibilities for two-reading certainty, including randomized policies (Proposition 2). Their optimal success probabilities below one are assessed separately. The fixed-budget calculation in Section 2.5.3 supplies an explicit fallback when two readings cannot guarantee success. Bulk is preferred on the measured public API when its full response is admissible (Section 3.4).

The central result is a boundary between one comparison and a shared collection of comparisons. For one affine contrast, a deterministic policy that guarantees precision with two exact scalar readings implies a randomized fixed batch with worst-case success at least one half (Theorem 1). With two contrasts, that floor disappears. For every $m \geq 3$, Theorem 2 constructs instances with optimal adaptive success 1 and optimal batch success $1/m$, still using two readings. Each contrast separately has a successful fixed batch; all coordinates are queryable at equal cost. The same separation holds for all pairs among three nonnegative forecast vectors (Corollary 1). The dictionary grows with m, and three fixed queries suffice. Moreover, the optimal batch mixture is unique: it assigns mass $1/m$ to each of m necessary batches. Even positive worst-record success requires all m batches. This exact support requirement isolates what fails when separately sufficient measurements share a budget.

We make that distinction computable for cumulative-meter audits. An exact two-query planner,

minimum fixed-budget calculation and matching randomized-batch certificates determine when feedback saves a reading. The study uses frozen forecasts on 1,098 blocks from six systems. Separate scalar-budget calculations isolate the shared-budget mechanism; absolute precision, nearby thresholds and capacity controls test its sensitivity on the same records. Replay then measures what the policies achieve on stored answers. A public-API control distinguishes the mathematical reading budget from the available retrieval interface.

The same calculation can rule out an insufficient budget. Explicit adverse records show that randomizing an adaptive policy cannot restore certainty on any of the 583 blocks without a deterministic guarantee (Proposition 2). Thus the method supports a three-way decision: choose a fixed batch, use feedback, or obtain more information.

The paper develops the scalar bound, the shared-budget construction and the meter planner, then reports its certificates, recorded outcomes and controls. The retrospective study reuses frozen source panels. Online Resource 1 retains the supporting scalar penalty classification (S11), a separate mean-joint-width construction (S14), historical controls and executable source-to-table verification.

1.1 Related work and the contribution boundary

Partial identification studies conclusions that survive incomplete observations (Manski, 2005; Maia Polo et al., 2024). Cancellation in paired squared losses already yields an affine function of the missing truth (Patton, 2011). Active evaluation and model selection acquire labels to estimate performance or choose predictors (Sawade et al., 2012; Hara et al., 2024; Kossen et al., 2021; Okanovic et al., 2025). We instead require the entire compatible interval of each fixed-record contrast to meet a declared precision target. Kuboyama (2026, Theorem 1 and Proposition 1) studies completion-based model-choice certificates with adaptive binary labels and shared paired bounds. Its distinction between hindsight certificate size and acquisition requirements (Theorem 2 and Proposition 3) is also relevant here; our guarantees concern joint widths for continuous targets. The width of an information fiber is the established local diameter of information (Novak and Woźniakowski, 2008, Section 4.1). Our shared objective is its weighted ℓ_∞ version for a vector target. Classical deterministic adaptivity bounds cover linear information on convex symmetric classes (Novak and Woźniakowski, 2008, Theorems 4.4–4.5); nonsymmetric recovery is studied by Novak (1995), and noisy scalar criteria by Osipenko (2019, Theorems 2–3 and 5). Modern randomized recovery minimizes reconstruction error (Krieg et al., 2025, Section 2 and Proposition 4.1) (Krieg and Ullrich, 2026, Section 9.1, Eq. (9.15), author manuscript v2). Adaptive vector reconstruction also admits high-probability guarantees and growing gains over nonadaptive reconstruction (Kunsch and Wnuk, 2026, Theorems 3.1 and 4.1, author manuscript v1). Our precision objective requires the full terminal fiber to be narrow. The shared-budget construction keeps the output dimension and hard query budget at two as the predeclared coordinate dictionary grows. For supporting mean-width results, the order of the worst-record and randomization operations also differs from expected global radius (Krieg and Ullrich, 2026, Eq. (2.4), Remark 6.1, author manuscript v2). Online Resource 1, S11 gives that comparison and the scalar penalty classification, whose sufficient direction applies Sion's minimax theorem (Sion, 1958, Theorem 3.4) and Jensen's inequality.

Information-dependent decisions are established in adjustable robust optimization (Wei and Zhang, 2024). Vayanos et al. (2026, Section 3.2, Eq. (3) and Theorem 3.2, author manuscript v3)

give equivalent observation-consistent recourse and information-fiber formulations. Our scalar precision bound adapts the sufficient-policy result of Kurtz (2026, Theorem 2 and Example 5). Kurtz's Theorem 11 also permits policy counts to grow with recourse-region complexity. His Example 20 gives a particularly close construction: a scalar uncertainty coordinate selects an interval on which a recourse policy is feasible, and the chosen policies must cover those intervals. Our construction shares this interval-cover mechanism. The difference is how it arises: we derive the intervals from the sharp widths of actual coordinate-observation fibers, with every coordinate available at the same cost. Each scalar conditional width is affine, but their joint maximum loses the concavity used by the scalar bound. Two affine targets and two unit-cost readings have exact success values 1 and $1/m$, although each target alone has a sufficient fixed batch. The adverse records further force support m for any positive batch success and uniquely determine the optimal mixture. The output dimension and budget stay fixed while the measurement dictionary grows. This is the specific contribution beyond general policy-count bounds and the established benefits of adaptive vector recovery.

Adaptive cumulative sampling has close precedents. Gaillard et al. (2023, Section 4.1.1, Corollary 1 and Theorem 3) studies certified scalar integral estimation from a monotone function. For monotone contrast coefficients, summation by parts relates our meter problem to weighted monotone integration. We pair an exact shared-query planner with continuum covering and adverse-record certificates, so each reported precision separation has matching bounds under one measurement interface. The separate aggregate solver in Online Resource 1, S5/S7 builds on weighted-oracle randomization (Kawase and Sumita, 2024), dynamic-program LP representations (Martin et al., 1990), and refinement of tied robust solutions (Iancu and Trichakis, 2014); it is supporting computation rather than the central shared-query contribution.

2 Methods

2.1 Model Comparisons and Sharp Information

Let $a, b \in \mathbb{R}^N$ be two fixed prediction vectors for the same scheduled targets. Their paired mean-squared-error difference is

$$D(y) = \frac{1}{N}\sum_{i=1}^{N}\{(a_i - y_i)^2 - (b_i - y_i)^2\} = d + c^\top y, \qquad c_i = \frac{2(b_i - a_i)}{N}. \tag{1}$$

All scheduled targets remain in the denominator. Initially observed values, documented aggregate measurements and other defensible constraints define the feasible set F. Any permitted archive record that directly reconstructs a target is incorporated before this set is declared. Sharpness is always relative to that declared information set.

The attainable interval of D on F is an identification interval for this record, not a sampling confidence interval. A negative upper endpoint certifies that a has smaller squared loss for every completion; a positive lower endpoint certifies b. An interval containing zero cannot certify a strict winner. Subsequent exact measurements restrict F to a fiber and therefore cannot widen its sharp interval. A prescribed width threshold instead asks for a precision guarantee, regardless of the interval's location.

The mathematical examples below also describe paired squared-loss widths. Given any contrast coefficient c, nonnegative predictions $a_i = N\max(-c_i, 0)/2$ and $b_i = N\max(c_i, 0)/2$ realize that coefficient in (1). The induced constant changes the interval's position while preserving its width. This gives the constructions a paired model-comparison interpretation.

2.2 Information and precision

Let $F \subset \mathbb{R}^n$ be nonempty, compact and convex, and let $T(y) = c^\top y + d$ be the scalar contrast. We select exact linear measurements from a fixed finite dictionary. The nonempty family $\mathcal{Q}$ of legal final query sets is fixed before observing answers and is the same for batch and adaptive policies. For example, fixed costs $k_j \geq 0$ and a hard budget B require $\sum_{j\in Q} k_j \leq B$ on every execution. Write L_Q for the matrix of measurements in Q. The dictionary specifies available measurements; it need not contain the target contrasts. Allowing direct queries of two targets within the budget would give a fixed solution with zero width. In the meter application, available queries return cumulative readings.

For a feasible record y, the compatible records and sharp width are

$$F_Q(y) = \{z \in F : L_Q z = L_Q y\}, \qquad W_Q(y) = \max_{z\in F_Q(y)} T(z) - \min_{z\in F_Q(y)} T(z). \tag{2}$$

Compactness guarantees both endpoints. This width is the scalar local diameter of information; additive constants in T do not affect it. A randomized batch chooses Q before seeing answers. An adaptive policy uses previous answers and an independent random seed to choose later measurements and when to stop. Query and stopping rules are jointly measurable, and every execution ends after finitely many queries in $\mathcal{Q}$. The hidden record is fixed before the seed is drawn. For bounded-error models, one fixed error vector is part of that record throughout the execution; the resulting feasible set must remain convex.

For either randomized policy class $\mathcal{C}$, define precision success by

$$S_\tau^{\mathcal{C}} = \sup_{\mathcal{P}\in\mathcal{C}} \inf_{y\in F} \Pr_{\omega}\{W_{Q_{\mathcal{P}}(y,\omega)}(y) \leq \tau\}, \qquad \tau \geq 0. \tag{3}$$

Probability concerns the policy's seed at a fixed record; equality at the threshold counts as success.

Proposition 1 (Width and the transcript fiber)**.** *For fixed Q, W_Q is finite, nonnegative, concave and upper semicontinuous on F. For a fixed policy seed, the records consistent with an adaptive transcript are exactly $F_Q(y)$ for its terminal set Q.*

Proof. Convex combinations of maximizing/minimizing endpoint pairs are feasible at the corresponding combined record, proving concavity. Along a convergent sequence of records, compactness supplies convergent endpoint subsequences; their limits are feasible in the limiting fiber, proving upper semicontinuity. Thus the widths are Borel measurable. For a fixed seed, records agreeing on queried answers produce the same next query and stopping decision, by induction. Conversely, a consistent transcript requires those answers. The argument includes a transcript with no queries. □

2.3 The scalar two-query precision floor

The following scalar specialization adapts sparse-policy reasoning (Kurtz, 2026, Theorem 2 and Example 5). Online Resource 1, S12 gives the general first-answer-dimension bound and its proof.

Theorem 1 (A half-success guarantee for scalar precision)**.** *Under Section 2.2, suppose a deterministic policy using at most two exact scalar measurements guarantees $W \le \tau$ for every record. The eligible second measurements are fixed before the first answer, and every resulting final set obeys the common hard budget. Then a randomized nonadaptive policy supported on at most two legal batches has worst-case precision success at least $1/2$.*

Proof. If the policy makes no query, the empty batch succeeds. Otherwise its first measurement $L_0 y$ is fixed. Its feasible answers form a compact interval $I = L_0 F$, possibly a singleton. Index the finitely many legal second actions by j, including an empty action when stopping after the first query is legal. Let

$$g_j(t) = \max\{c^\top (u - v) : u, v \in F,\ L_0 u = L_0 v = t,\ L_j u = L_j v\}.$$

This is the largest terminal width after choosing action j at first answer t: each feasible endpoint pair belongs to a common terminal fiber, and every such fiber supplies feasible endpoints. Convex combinations of endpoint pairs show that g_j is concave. Hence its strict failure set $H_j = \{t \in I : g_j(t) > \tau\}$ is convex. The policy's guarantee implies $\bigcap_j H_j = \varnothing$. Finite Helly's theorem in dimension at most one supplies at most two failure sets with empty intersection (Amenta et al., 2016, Theorem 1.1). This form does not require closed sets, so equality at the precision threshold is preserved. Choose uniformly among the corresponding fixed batches (L_0, L_j). At every record, at least one selected batch has width at most $g_j(L_0 y) \le \tau$. Its success probability is therefore at least $1/2$ (or one if a single batch suffices). □

2.4 A fixed-budget boundary between scalar and joint precision

The scalar half-success floor need not survive adding a second target. The following construction keeps the query budget at two and makes every coordinate available at unit cost.

Theorem 2 (Two-target precision with a fixed two-query budget)**.** *For every integer $m \ge 3$, let*

$$F_m = \{(t, u, v, z_1, \dots, z_m) : 0 \le u \le t \le 1,\ 0 \le v \le 1 - t,\ 0 \le z_j \le 1\}.$$

All $m + 3$ coordinates are exact unit-cost queries. Define

$$a_j = j + \tfrac{1}{4}, \quad b_j = m + 1 - j + \tfrac{1}{4}, \quad H = \frac{m(2m+3)}{4}, \quad M = H - m,$$

and the affine targets

$$T_1 = Mt + mu + \sum_j a_j z_j, \qquad T_2 = M(1 - t) + mv + \sum_j b_j z_j.$$

Both initial target intervals are $[0, 2H]$. With at most two queries on every execution, let success mean that both terminal sharp widths are at most H. Then the optimal worst-record precision-success probabilities are

$$S^{\star}_{\text{adaptive}} = 1, \qquad S^{\star}_{\text{randomized batch}} = \frac{1}{m}.$$

Each scalar target separately admits a fixed two-query policy with sure success. The adversarial record is fixed before any policy randomization.

Proof. The polytope F_m is nonempty, compact and convex. Since $\sum_j a_j = \sum_j b_j = H$ and $M + m = H$, the extrema of T_1 are 0 and $2H$, attained at $(t, u, z) = (0, 0, 0)$ and $(1, 1, \mathbf{1})$. The corresponding choices $(t, v, z) = (1, 0, 0)$ and $(0, 1, \mathbf{1})$ give the same extrema for T_2.
Query t, choose the smallest index j satisfying $(j-1)/m \le t \le j/m$, and query z_j. The exact terminal widths are

$$W_1 = H + mt - a_j, \qquad W_2 = H + m(1-t) - b_j. \tag{4}$$

Indeed, conditioning on t leaves $u \in [0, t]$ and $v \in [0, 1-t]$; the unqueried z coordinates remain an independent box. The observed value of z_j contributes only a constant. The chosen branch gives $\max(W_1, W_2) \le H - 1/4$, proving adaptive success one.
For the randomized-batch lower bound, choose $j \in \{1, \dots, m\}$ uniformly before any observations and query $\{t, z_j\}$. Every feasible t belongs to at least one branch interval, so success probability is at least $1/m$ for every record. This lower bound holds over the full continuum of F_m.
For the upper bound, give equal adverse mass to the m feasible records

$$x_i = \big((i - \tfrac{1}{2})/m, 0, 0, 0, \dots, 0\big), \qquad i = 1, \dots, m,$$

and write $t_i = (i - 1/2)/m$. The following table lists the exact terminal widths on these records for every type of distinct two-coordinate batch:

queried coordinates	W_1	W_2
$\{t, z_j\}$	$H + mt_i - a_j$	$H + m(1-t_i) - b_j$
$\{t, u\}$	H	$H + m(1-t_i)$
$\{t, v\}$	$H + mt_i$	H
$\{u, v\}$	$H + M$	$H + M$
$\{u, z_j\}$	$H + M - a_j$	$2H - b_j$
$\{v, z_j\}$	$2H - a_j$	$H + M - b_j$
$\{z_j, z_k\},\ j \ne k$	$2H - a_j - a_k$	$2H - b_j - b_k$

For the first row,

$$\max(W_1, W_2) = H + |i - j| - \tfrac{3}{4},$$

so the batch succeeds exactly at x_j. The next two rows fail because $0 < t_i < 1$, and the fourth fails because $M > 0$. For the fifth and sixth, $a_j, b_j \le m + 1/4 < H$, so the displayed $2H$ term exceeds H. Finally,

$$(a_j + a_k) + (b_j + b_k) = 2m + 3 < 2H \qquad (m \ge 3).$$

At least one of $a_j + a_k, b_j + b_k$ is therefore below H, so every two-z batch fails as well. Thus each deterministic batch succeeds on at most $1/m$ of the adverse mass. The same holds for any batch mixture; its minimum success over records cannot exceed this adverse average.
Fewer than two distinct queries, repeated queries, and stopping early do not evade the bound. Extend any such query set to a distinct two-query set. Additional observations cannot enlarge either sharp interval, so the original success set is contained in the extended success set. The table therefore also bounds every permitted shorter execution. This completes the matching upper and lower bounds.
Finally, the fixed batch $\{t, z_m\}$ makes $W_1 \le H - 1/4$ for every record, and $\{t, z_1\}$ makes $W_2 \le H - 1/4$. The difficulty concerns serving both targets with the same two queries, not either target alone. □

Corollary 1 (All pairs of three models, threshold margin, and query cost)**.** *The construction can be realized as all three paired squared-loss comparisons among three nonnegative prediction vectors. The exact success values remain* 1 *and* $1/m$ *at every common normalized threshold*

$$\tau \in \left[\frac{1}{2} - \frac{1}{8H}, \frac{1}{2} + \frac{1}{8H}\right).$$

Each target separately remains fixed-batch certifiable, and three fixed queries suffice jointly throughout this interval. Hence the unbounded ratio concerns success probabilities at a common two-query budget, not an unbounded reduction in the number of readings.

Proof sketch. Add the third contrast $T_3 = T_1 - T_2$. Querying t alone makes its normalized width smaller than the lower endpoint of the stated threshold interval. The first two contrasts retain their adaptive and uniform-batch guarantees; the theorem's adverse records retain the matching batch upper bound. The fixed batch $\{t, z_1, z_m\}$ certifies all three. Explicit nonnegative prediction vectors and the threshold inequalities appear in Online Resource 1, S17. □

The adaptive normalized margin at half-width is $1/(8H) = 1/[2m(2m+3)]$, which shrinks with m. These are exact-query constructed instances, not a fixed-error robustness result or a new empirical advantage on the cumulative-meter data. The scalar sparse-cover theorem concerns one affine target. Here each conditional joint width is the maximum of two affine functions; its strict failure set can be disconnected, so the scalar Helly argument does not supply a half-success floor.

The optimal mixture is unique. In the family of Theorem 2, any randomized batch with positive worst-record success must assign positive probability to every batch $\{t, z_j\}$, $j = 1, \ldots, m$. The unique optimal distribution over preselected unordered batches assigns probability $1/m$ to each of these m batches. Both statements also hold for all three model comparisons and every threshold in Corollary 1. To see this, at the adverse record x_i, only $\{t, z_i\}$ succeeds. Thus success there equals its mixture probability p_i. A positive worst-record value requires every $p_i > 0$; achieving $1/m$ requires every $p_i \ge 1/m$. Since the probabilities sum to at most one, all equal $1/m$ and no mass remains for other sets. The uniform mixture already provides the matching guarantee on every feasible record. Repeated or fewer queries cannot succeed at any x_i. Early stopping cannot improve the success of a preselected batch, so it does not evade the necessary support or mixture weights. Thus fixed output dimension two and a scalar first answer do not impose a universal bound on the number of batches needed for positive joint assurance.

Executable checks. Online Resource 1, S17 supplies the rational verifier, a separately assembled LP check, numerical tolerances and exact input/output hashes. The complete proof of Theorem 2 above establishes its universal claim; these finite calculations check the implementation.

2.5 Shared precision from cumulative-meter readings

Consider a six-hour block with nonnegative hourly energy increments $z = (z_1, \dots, z_6)$ and retained cumulative readings C_0, C_6. In common energy units, $E = C_6 - C_0$ gives $F_E = \{z \geq 0 : \sum_h z_h = E\}$. For two frozen forecast vectors, the block MSE contrast is $D(z) = d + c^\top z$, with $c_h = 2(b_h - a_h)/6$. All six targets remain in its denominator.
The five optional queries return internal cumulative readings C_k, $k = 1, \dots, 5$, equivalently exact prefix sums $s_k = C_k - C_0 = \sum_{h \leq k} z_h$. The budget is two distinct readings shared by every model comparison in the block. The endpoints are initially available and do not use this optional-query budget.
For $0 \leq p < q \leq 6$, put $r(p, q) = \max_{p<h\leq q} c_h - \min_{p<h\leq q} c_h$. Order the queried and outer boundaries as $0 = q_0 < \cdots < q_m = 6$, with $s_0 = 0$ and $s_6 = E$. The exact terminal width is

$$W_Q(z) = \sum_{\ell=1}^{m} (s_{q_\ell} - s_{q_{\ell-1}}) r(q_{\ell-1}, q_\ell). \tag{5}$$

Within each segment, its nonnegative energy can be placed at an hour with the smallest or largest coefficient. These choices attain both endpoints, proving sharpness. Initially $W_0 = Er(0, 6)$; zero-width contrasts require no further information. Online Resource 1, S6 retains the separate scalar closed forms and their controls. An auditor comparing several models can reuse each reading across their loss contrasts. Let $W_{r,Q}(y)$ be the sharp width for contrast r after the common transcript Q, and let $W_{r,0} > 0$ be its initial width. Define

$$J_Q(y) = \max_r W_{r,Q}(y)/W_{r,0}.$$

Then $J_Q(y) \leq 1/2$ means that every requested comparison has reached half its original width. Already determined contrasts can be omitted. This is the local diameter of the vector contrast map in a weighted ℓ_∞ norm, within the established vector-recovery framework (Krieg et al., 2025); shared-label model certification also has close prior work (Kuboyama, 2026).

2.5.1 Planning two shared prefix queries

We plan one pair of readings for all requested contrasts. The inputs are m normalized coefficient rows $c_r \in \mathbb{R}^n$, each with range one, an exact total $E > 0$, and all internal prefix queries $P_k x = \sum_{h \leq k} x_h$, where $x = y/E \geq 0$ and $\sum_h x_h = 1$. We assume $n \geq 3$ and a budget of two distinct queries. Define the range over a consecutive segment by $\Delta_r(a{:}b) = \max_{a\leq h\leq b} c_{rh} - \min_{a\leq h\leq b} c_{rh}$.
Suppose the first query is k and its answer is t. For a proposed second query $j \neq k$, the conditional worst width of contrast r is $\alpha_{kjr} t + \beta_{kjr}(1 - t)$, where

$$(\alpha_{kjr}, \beta_{kjr}) = \begin{cases} (\max\{\Delta_r(1{:}j), \Delta_r(j+1{:}k)\}, \Delta_r(k+1{:}n)), & j < k, \\ (\Delta_r(1{:}k), \max\{\Delta_r(k+1{:}j), \Delta_r(j+1{:}n)\}), & j > k. \end{cases}$$

Within each side of the first boundary, a worst-case allocation places its mass in the segment with the larger coefficient range. Maximizing over records and over the finite contrast family commutes, so

$$g_{kj}(t) = \max_r \{\alpha_{kjr} t + \beta_{kjr}(1-t)\}, \qquad A = \min_k \max_{t\in[0,1]} \min_{j\neq k} g_{kj}(t)$$

is the optimal deterministic two-query worst-width guarantee.

Algorithm: two shared queries, exact-total simplex

1. For each first boundary k, form every affine line $\ell_{kjr}(t) = (\alpha_{kjr} - \beta_{kjr})t + \beta_{kjr}$.
2. Form B_k from 0, 1, and every intersection in $[0, 1]$ of two nonparallel lines, including lines belonging to different second queries.
3. Compute $A_k = \max_{t\in B_k} \min_{j\neq k} \max_r \ell_{kjr}(t)$. Choose the smallest boundary index attaining $\min_k A_k$.
4. Retrieve that boundary's counter value and calculate its normalized prefix mass t using the retained endpoint total.
5. Choose the smallest $j \neq k$ attaining $\min_j g_{kj}(t)$, retrieve it, and calculate all terminal segment-width certificates using the two shared answers.

Between consecutive points of B_k, the ordering of all underlying lines is fixed. The maximum within each group and then the minimum over groups therefore select one affine line on that interval. Its maximum is at an endpoint. This proves that the finite calculation covers every possible first answer; it is not a grid approximation. Equal or coincident lines are handled by the stated index rule.

The mathematical algorithm uses exact comparisons. The reported six-interval implementation uses the same construction in floating-point arithmetic: it treats values within 10^{-12} of a minimum as tied and chooses the smallest index, both offline and online. Its intersection routine skips slope differences of magnitude at most 10^{-14}; a 257-point grid is an additional numerical check. Exact rational branch and adversarial certificates support the reported exact assurance claims. The algorithm above addresses deterministic worst width on the uncapped, exact-total simplex. Bounded-error and capacity analyses use the separate conditional calculations described in their sensitivity protocols; randomized mean-width and precision-success comparisons retain their own optimization objectives.

2.5.2 Certifying the optimal randomized batch

The ten two-reading batches define a small precision game. An at-most-two-reading policy can be padded to two readings without reducing its precision success, so this list also covers smaller batches. For batch q, let v_{qr} assign the normalized coefficient range of contrast r to every hour in each segment. Its success set is

$$S_q = \{x \geq 0 : \mathbf{1}^\top x = 1,\ v_{qr}^\top x \leq \tau \text{ for every } r\}.$$

A mixture π has precision success $\sum_q \pi_q \mathbf{1}\{x \in S_q\}$ at record x. We compute its optimal worst-record value by alternating a finite game with a full-domain check.

The finite game returns a candidate mixture and an adverse distribution over feasible records. To verify a proposed lower guarantee s, consider any set M of failed batches with $\sum_{q \in M} \pi_q > 1 - s$. Such a failure set must be impossible. We check it by branching over a failing contrast for each batch: failure requires $v_{q,r_q}^\top x > \tau$. For each nonempty branch, we maximize a common slack δ subject to $x \geq 0$, $\mathbf{1}^\top x = 1$ and $v_{q,r_q}^\top x \geq \tau + \delta$ for every selected row. A positive optimum characterizes strict feasibility; we accept proposed witnesses only after exact rational verification. The LP proposes either such a record or nonnegative rational weights satisfying

$$\sum_{q \in M'} \lambda_q = 1, \qquad \sum_{q \in M'} \lambda_q v_{q,r_q,h} \leq \tau \quad (h = 1, \ldots, 6)$$

for the contrasts selected along the current branch $M' \subseteq M$. The latter inequalities contradict simultaneous strict failure on the simplex and close that branch. Within each batch, we omit a contrast row only if another retained row dominates it componentwise, so every possible failure remains covered. A feasible failure record is added to the game, and we repeat.

The resulting proof consists of branch covers for the lower bound and an adverse-record distribution for the upper bound. A solver-free checker reconstructs the coefficients, verifies the rational inequalities, and checks all $2^{10} = 1{,}024$ possible failure masks. Matching bounds establish the continuum optimum. This computation uses forecasts and initial totals; optional reading answers enter only the subsequent replay.

Choosing among precision-optimal mixtures. We freeze the optimal worst-record precision value before choosing the replay policy. Among mixtures with that value, we numerically minimize worst expected joint width. This secondary objective needs only the six simplex vertices because each batch's joint width is convex in the record. We resolve remaining ties lexicographically by batch index, allowing 10^{-10} in the secondary width objective, rationalize the probabilities, and recheck the primary precision guarantee over the full simplex. When the precision optimum is zero, every mixture is precision-optimal, so we retain the saved coefficient-only mean-width policy. No recorded reading enters this selection. Primary precision optimality is exact; the secondary width refinement uses numerical LP solutions.

2.5.3 Computing the minimum fixed reading budget

We can choose the reading budget instead of assuming it. Here $F = \{x \geq 0 : \sum_h x_h = 1\}$ and $\tau \geq 0$. For the normalized coefficients above, a fixed prefix set Q partitions the record into segments $\mathcal{S}(Q)$. Concentrating the unknown mass in one segment gives

$$\sup_{x \in F} J_Q(x) = \max_{r, S \in \mathcal{S}(Q)} \operatorname{range}(c_r|_S), \qquad b_{\text{fix}}(\tau) = \min_Q \{|Q| : \sup_{x \in F} J_Q(x) \leq \tau\}.$$

This yields a simple exact algorithm. Starting at the first interval, extend the current segment while every coefficient row has range at most τ. When the next interval would violate a bound, place a query immediately before it and start a new segment. The resulting partition has the fewest segments, hence the fewest readings: its first segment ends no earlier than that of any optimal

partition, whose remaining segments can be truncated to give a valid partition of the remaining suffix. Induction completes the exchange argument. Maintaining each row's minimum and maximum takes $O(Rn)$ comparisons for R contrasts and n intervals.

Randomization cannot reduce this fixed budget when success one is required. Any positive-probability batch that fails on a feasible record prevents certainty at that record. We compare the fixed minimum with the existing two-query adaptive certificate. If no single reading suffices and that certificate exists, the adaptive minimum is exactly two. Otherwise the minimum fixed batch supplies a constructive fallback; we do not optimize adaptive budgets above two. This calculation concerns exact, uncapped nonnegative allocations and noiseless prefix readings.

Separating shared-budget difficulty from absolute loss scale. We compute the minimum fixed budget for each contrast separately and for the six contrasts together. A block has a *joint-only obstruction* when every scalar contrast needs at most two fixed readings, but their joint requirement needs three. We also enumerate contrast subsets to identify the smallest number that causes this obstruction. A contrast is individually binding when deleting it lowers the joint minimum fixed budget; redundant constraints can make several contrasts collectively binding without any single one passing this deletion test.

We recorded an absolute-floor sensitivity protocol before computing its outcomes. For each system, let M_2 be mean squared hourly consumption over the days used for its frozen model fits. We replace the relative requirement by

$$W_r \leq \max\{W_{r,0}/2, \epsilon\}, \qquad \epsilon = \gamma M_2,$$

using $\gamma = 0.01$ as the primary sensitivity and $\gamma \in \{0, 0.001, 0.005, 0.01, 0.025, 0.05, 0.1\}$ as the full grid. Both sides have squared-kWh units. We reconstruct M_2 from training counter differences with the original complete-day and lag-1,2,7 eligibility rules, and verify the frozen training means. No withheld evaluation answer determines the floor. The fixed-budget and two-query algorithms use the resulting contrast-specific thresholds $\max\{1/2, \epsilon/W_{r,0}\}$. This is a retrospective sensitivity analysis of precision requirements.

2.5.4 When two readings cannot guarantee precision

The planner also identifies an insufficient budget. Randomizing the adaptive choices cannot restore certainty in this case. The following finite-witness argument makes that conclusion checkable.

Proposition 2 (A certificate against randomized adaptive certainty)**.** *In the six-hour exact model with five permitted internal prefixes, suppose $A > \tau$. Then every randomized adaptive policy using at most two readings has worst-record precision success at most $19/20$. The bound need not be tight. Thus a policy with success one exists if and only if a deterministic adaptive policy with success one exists.*

Proof. For each of the five first queries k, choose an answer t_k at which every second query $j \neq k$ has $g_{kj}(t_k) > \tau$. Such an answer exists because $A > \tau$. For each ordered pair (k, j) choose a feasible record x_{kj} with first answer t_k on which the terminal sharp joint width exceeds τ. Put

mass $1/20$ on these twenty records, retaining multiplicities. This adverse distribution is fixed before the policy's independent random seed is drawn.

Fix a seed. It determines a first query k and, upon answer t_k, a second query j. All four records in row k have that same first answer, so the policy follows this path on x_{kj} and fails. Its failure probability under the fixed adverse distribution is therefore at least $1/20$. Averaging over seeds preserves the bound. For each randomized policy there is therefore a fixed record with success at most $19/20$. Early stopping or repeated queries can be padded with unused legal queries; failure after padding implies failure before it because observations only shrink the compatible intervals. Conversely, a deterministic successful policy is a randomized policy with a constant seed. This proves the claim. □

For each x_{kj} the executable certificate supplies two nonnegative, unit-sum endpoint records with identical queried answers and a contrast difference strictly above τ. One endpoint serves as x_{kj}; the other proves failure of its sharp interval. The verifier checks these equalities and inequalities directly, without using an optimizer's verdict. This elementary finite-action argument concerns certainty under the stated model; it does not compute the optimal success probability below one. Figure 2 shows the conditional curves and the selected second query on a recorded task.

2.5.5 A worked external certificate

Consider HPM 1 on 5 July 2025, 12:00–18:00 UTC: the first matched-half case in the frozen external ledger. Its retained endpoints give $E = 1.83984375$ kWh. Normalize the six model-pair coefficient ranges to one and write $x = y/E$, so $x \geq 0$ and $\sum_h x_h = 1$.

The policy first queries boundary 2. It then queries boundary 4 when the answer $t = (C_2 - C_0)/E$ is below $\theta \in [0.5226, 0.5227]$, and boundary 1 above it. The two branch bounds are 0.47743079 and 0.4774; either action is safe within the switch enclosure. Both bounds are below the required $J \leq 1/2$. Figure 2 shows the exact envelope, whose peak is about 0.47734.

For fixed batches, mix $Q = \{1, 2\}$ and $R = \{2, 4\}$ equally. A rational cover certificate shows that these batches cannot both fail on any feasible record, giving success at least $1/2$. Conversely, put half the adverse mass at e_2 and half at $(e_4 + e_5)/2$. Batch Q fails on the second record; every other two-reading batch fails on the first. Each batch therefore succeeds on at most half the adverse mass. The batch optimum is exactly $1/2$, while adaptation succeeds on every completion.

The initial six-hour MSE-difference widths range from 0.107125 to $0.303143\,\text{kWh}^2$; their half-width targets range from 0.053563 to $0.151571\,\text{kWh}^2$. Online Resource 1, S17 supplies the full coefficient enclosures, 36 cover inequalities, adverse-record checks, absolute-width table and a solver-free verifier.

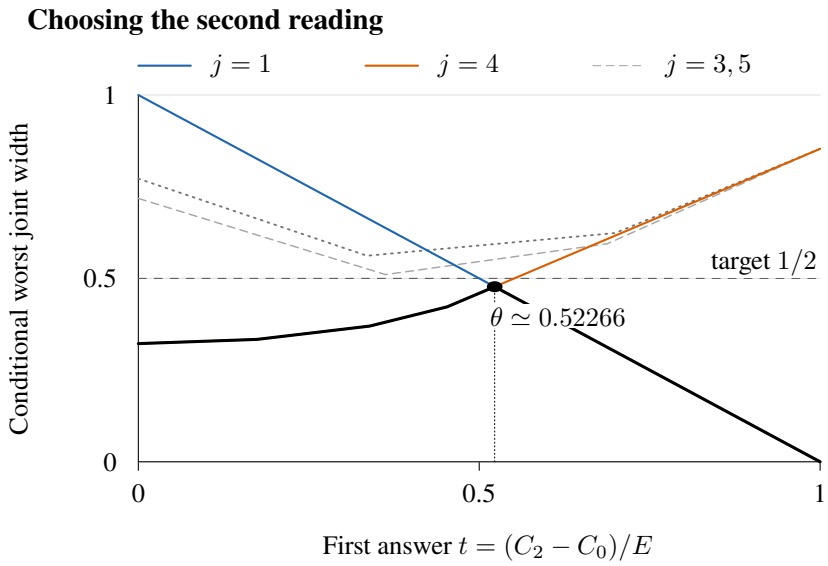


Figure 2: Choosing the second reading in the HPM 1 worked case. The black lower envelope selects boundary 4 below the switch and boundary 1 above it. Its maximum, about 0.47734, is below the half-width target for every first answer. Gray curves show the other permitted second readings.

2.6 Source panels and audit protocol

We use a prespecified three-consumer case series: the lexicographically first three IDs in the pinned BLEM cumulative-counter source (Kostmann and Härdle, 2019). The source-preparation protocol fixed this ID rule before model fitting and evaluation. Training covers January–September 2017 and testing October–December 2017. Complete-day and feature filters leave 91 test days per meter. Hourly energy uses integer counter differences at exact UTC endpoints, converted to kWh without interpolation; partial days are excluded. Online Resource 1, S3 records source hashes and fitting details.

The external panel uses HeatpumpMonitor systems 1, 2 and 7 (OpenEnergyMonitor, 2026a,b). We screened ascending IDs for public MID status, Class-1 electrical-meter metadata and cumulative coverage of January–July 2025 before hourly retrieval. Training-only gates, fixed before July retrieval, required 95% valid hourly intervals and 90 complete training days. All three passed without replacement. January–June supplied training; July supplied 31 test days per system. We convert stored API-grid increments to kWh. Native acquisition and historical delivery times are unavailable (OpenEnergyMonitor, 2026c); S6 records the source-selection chronology. Both panels reuse frozen daily-persistence, ridge, boosted-tree and MLP forecasts. Each external model specification was fitted once without retuning. We audit blocks ending at 06:00, 12:00 and 18:00 UTC, giving 819 original and 279 external blocks. At each declared cutoff, all policies receive the endpoints and day-start forecasts; the following day's persistence copies are not yet available under this benchmark schedule. Two internal readings serve all six pairs of the four models. This is a benchmark access contract, not a verified restriction of either source's service; the public-API experiment also evaluates unrestricted bulk retrieval. The shared-query analysis is retrospective on both panels. Policies use frozen forecasts and, for adaptation, previously queried

answers. We open withheld readings only for replay and require $W_r \le W_{r,0}/2$ for every pair, using the uncapped exact-total initial widths throughout the shared-query sensitivity controls.

What the measurement model certifies. Table 1 defines the stored-record and bounded-error models used below.

Table 1: Targets and observation assumptions in the application. Error allowances and hourly caps are sensitivity inputs, not verified device limits.

Model	Target and observations	Scope of the guarantee
Exact stored samples	Targets are increments of stored cumulative samples. Retained endpoints and queried internal samples are exact in this record model.	Loss-contrast intervals on the stored sample grid.
Bounded internal errors	The latent record includes one fixed internal-error vector with $\lvert e_k\rvert \le \delta E$, where E is the retained block total. Endpoints, and hence the retained total, remain exact. Errors are fixed before policy randomization.	All records and fixed error vectors within these bounds.
Physical energy	Actual energy and meter readings would require defensible bounds for internal measurements, endpoints and any physical capacity constraints.	We provide no verified physical calibration. Source meter-class metadata alone does not establish the pointwise error model or exact endpoints.

Unchanged precision under capacity controls. We retain the uncapped exact-total initial width $W_{r,0}$ and the target $W_r \le W_{r,0}/2$, so every shared capacity control tests the same absolute uncertainty budget. The scalar control in Online Resource 1, S6 also reports results relative to the capped initial widths.

3 Results

We report precision guarantees, recorded outcomes and the reading budgets needed to meet the half-width target. All methods use the same initial information and permitted queries. The exact-reading panel divides into three acquisition decisions:

$$1{,}098 = \underbrace{290}_{\text{fixed two suffice}} + \underbrace{225}_{\text{feedback saves one}} + \underbrace{583}_{\text{more than two required}} .$$

Within the 225 reading-saving blocks, 191 isolate the joint-only mechanism. Of the 222 matched-half certificates, 179 remain valid throughout $[0.49, 0.51]$. The following analyses use these fixed subsets.

Table 2: Numerical conventions for the central shared-query analysis. Exact certificates and floating-point replay have distinct roles.

Calculation	Convention
Constructed and matched certificates	Rational inequalities on the supplied exact inputs; no outcome tolerance.
Planner ties and intersections	Treat values within 10^{-12} of a minimum as tied; use the smallest index. Skip slope differences of magnitude at most 10^{-14}.
Shared mean-batch LP	Primal and dual feasibility tolerances 10^{-9}; secondary selection allows primary objective slack 10^{-10}.
Full-panel floating-point replay	Classify normalized widths using $J \leq \tau + 10^{-8}$.
Exact assurance certificates	Rational inequalities use $J \leq \tau$. Recorded replay reports its strict-threshold sensitivity separately.
Supplementary coefficient enclosures (S17)	The 10^{-4} interval is a rational enclosure of displayed coefficients, not a solver feasibility tolerance. Full coefficients drive the checks.

3.1 Reading budgets and the shared-query mechanism

How many readings does assurance require? The minimum fixed budget is three on every block where two adaptive readings suffice but two fixed readings do not: 168 original and 57 external blocks, spanning all six systems. Thus two adaptive readings replace three necessary and sufficient fixed readings on 225 blocks, a one-third reduction. Each replacement saves one reading in exchange for an additional information round: the three fixed readings can be chosen together. The original count includes the 165 matched-half cases and three further blocks; only the matched cases establish exact batch success one half. Using adaptive two on these blocks and the minimum fixed batch elsewhere gives the complete-panel reading totals in Table 3.

Table 3: Reserved optional scalar readings for guaranteed joint half-width precision. Each plan succeeds for every completion in the declared exact model. Counts exclude the two common retained endpoints. The adaptive plan uses a fixed fallback where no two-reading guarantee is available.

Quantity	Original	External
Blocks	819	279
Three fixed versus two adaptive	168	57
Minimum fixed readings, summed	2625	743
Adaptive with fixed fallback, summed	2457	686
Reading reduction across full panel	6.40%	7.67%

We calculated these budgets from the frozen coefficients before using any optional reading answers. Enumeration of all 32 fixed subsets per block agrees with the linear scan. A separately implemented shortest-path checker verifies minimality and checks feasible endpoint witnesses and the adaptive interval covers. Online Resource 1 supplies each concrete fallback batch, source and system summaries, and the fixed-budget histogram. The capacity and error controls test how

additional information changes the acquisition requirement.

Which cases isolate the shared-query mechanism? Of the 225 two-versus-three reading separations, 191 are joint-only: 137 original and 54 external blocks have scalar minimum fixed budgets of at most two for every contrast. Within these 191 blocks, two contrasts suffice to cause the obstruction in 168, while 23 require at least three. The other 34 blocks contain a scalar contrast that already needs three fixed readings. Among the 222 matched-half certificates, 188 are joint-only.

3.2 Joint precision with two shared readings

Exact comparison with randomized batching. We solved the half-width precision game on every block. Table 4 gives the exact optima: matching feasible adverse distributions and rational branch covers close all 1,098 cases. On the 225 blocks where feedback saves a reading, adaptation guarantees success, whereas the best randomized batch has worst-record success $1/2$ on 222 blocks and $1/3$ on the remaining three. The comparison therefore includes every randomized mixture of the ten permitted batches.

Table 4: Exact optimal worst-record batch precision success with two readings. Each row partitions the complete panel; all lower and upper bounds match.

Panel	Blocks	0	$1/5$	$1/4$	$1/3$	$1/2$	1
Original	819	434	1	13	44	165	162
External	279	81	0	2	11	57	128

For the 418 of 1,098 games whose initial lower and upper precision bounds differed, exact completion took 92.09 seconds on the recorded host (maximum 2.27 seconds per case; 5,261 LPs). The independent checker verified 428,032 failure masks, 2,328 rational dual inequalities and 96,820 adverse-record/batch evaluations. These certificates are exact for the stored binary64-derived coefficients. The LPs propose witnesses; rational arithmetic determines whether they prove the reported bounds.

Shared-budget evaluation. We set the half-width target and plan the policies before opening the withheld register responses. The greedy control first minimizes the worst one-query joint width, breaks ties using the best two-query continuation, and then minimizes conditional worst width after the first answer.
The adaptive success gains over the best fixed batch for the worst-width objective are 7.81, 11.11 and 9.68 percentage points, respectively. Its gains over greedy are smaller: 2.20, 2.15 and 2.51 points. For the exact external panel, the mean realized J is 0.2823 for adaptive, 0.3381 for the fixed batch and 0.2893 for greedy. At the 1% allowance these are 0.3028, 0.3481 and 0.3099. The next comparison uses a batch policy designed for precision.

Replay of the precision-optimal batch. The randomized batch that optimizes worst-record precision has expected success over its frozen mixture of 54.15% on original and 71.83% on

Policy	Planning objective	Guarantee and recorded comparison
Exact adaptive	Minimize worst joint width	Optimal among deterministic two-query policies in the declared exact model; replay reports joint precision success.
Fixed batch	Minimize worst joint width	Best deterministic two-query batch for worst width.
Greedy adaptive	Myopic first query; optimal continuation	Chooses the first query myopically, then minimizes conditional worst width.
Randomized mean batch	Minimize worst expected joint width	Batch-optimal for the uncapped exact simplex; replay reports expected width and precision success separately.
Randomized precision batch	Maximize worst-record precision success	Globally optimal on all 1,098 blocks; coefficient-only width refinement selects among tied optima.

Table 5: Planning objectives under the same two-reading budget, information and query dictionary. Recorded precision uses the half-width target $J \leq 1/2$; each row states the corresponding optimization guarantee.

Panel / observation model	Blocks	Adaptive	Fixed batch	Greedy
Original, exact	819	641 (78.27%)	577 (70.45%)	623 (76.07%)
External, exact	279	252 (90.32%)	221 (79.21%)	246 (88.17%)
External, 1% allowance	279	242 (86.74%)	215 (77.06%)	235 (84.23%)

Table 6: Recorded-answer joint precision success with two shared readings. We select the fixed batch by its worst-width objective before answers; the last row reoptimizes this comparator for the error model while retaining the coefficient-derived adaptive and greedy policies. The 1% allowance enlarges compatible records around the stored answers with exact endpoints, as defined in Table 1.

external recorded blocks, compared with 78.27% and 90.32% for adaptation. The exact certificates establish the worst-case separation; these replay differences compare specified policies on the six observed systems. Online Resource 1, S18 gives the complete randomized comparison, frozen mixtures and descriptive intervals.

Capacity control. For an illustrative cap $y_h \leq E/2$, we reoptimize the noiseless planner using concave piecewise-linear segment widths. We solve each conditional problem by checking all line intersections within each capacity interval. Of the 819 original and 279 external records, respectively 780 and 200 satisfy this cap. Among those compatible records, adaptive-only half-width guarantees remain in 56 and 4 blocks; recorded adaptive versus fixed-batch successes are 655 versus 627 and 179 versus 177. Under the stronger cap $E/3$, only 613 original and 198 external records are compatible. No adaptive-only half-width guarantee remains, and the two planners have equal success counts of 613 and 192. We restrict cap-based guarantees to compatible records. Thus supplied capacity information can remove the need for feedback.

Combining the $E/2$ cap with the 1% stored-reading allowance, the capacity-aware frozen adaptive policy succeeds on 169/200 compatible records, versus 166/200 for the best fixed batch under that model. Mean joint widths are 0.3479 and 0.3535. The greedy control succeeds on 165/200 using its frozen uncapped policy; it has not been reoptimized for the capacity model.

Paired uncertainty and source variation. The external exact-panel advantage over the worst-width fixed batch is 11.11 percentage points (descriptive 95% interval 8.60–13.62), compared with 2.15 points over greedy (1.08–3.23). These percentile intervals use 2,000 seven-day circular calendar-block resamples, synchronized across the three systems and intraday blocks, with seed 20261013. The combined cap/error gain over the fixed batch is 1.50 points (-0.49–3.16). Paired 2×2 counts, individual-system effects and one-day-cluster sensitivity appear in Online Resource 1, S8. Table 7 shows the system-level paired counts. The external gain

System	n	A / F / G	A-only/F-only	Δ_F (pp)	A-only/G-only	Δ_G (pp)
O1	273	212/193/211	27/8	6.96	3/2	0.37
O2	273	196/187/188	30/21	3.30	9/1	2.93
O3	273	233/197/224	38/2	13.19	12/3	3.30
HPM 1	93	76/75/73	3/2	1.08	3/0	3.23
HPM 2	93	91/66/87	25/0	26.88	4/0	4.30
HPM 7	93	85/80/86	5/0	5.38	0/1	-1.08

Table 7: Exact-reading paired joint-precision outcomes by system. A is the exact deterministic adaptive planner; F is the best deterministic batch for worst width; G is greedy adaptation. The two discordant counts identify blocks where only A or only its comparator succeeds. O1–O3 are BLEM consumers 1–3; HPM labels retain provider system identifiers. Blocks are repeated observations within systems.

over fixed batching is heterogeneous: system 2 supplies 25 of the 31 net additional successes, while system 7 has one fewer success than greedy adaptation. Omitting each system in turn leaves positive exact-model panel effects: 5.13–10.07 and 3.23–16.13 percentage points over fixed batching for the original and external panels, respectively; the corresponding ranges over greedy are 1.65–3.11 and 1.08–3.76 points. A fourteen-day synchronized circular-block sensitivity also retains positive exact-model intervals (Online Resource 1, S8). For the combined cap/error model, the primary seven-day interval is $[-0.49, 3.16]$ points, the fourteen-day sensitivity is $[0.48, 2.63]$, and leave-one-system-out effects range from 0.81 to 2.63 points. The three-success gain is therefore sensitive to the temporal specification.

From precision to a model decision. We also ask whether the acquired readings certify a model within $\epsilon = 0.01 M_2$ of the best six-hour MSE, where M_2 is the training second moment. On the external panel, adaptive and greedy querying both certify 277 of 279 blocks, versus 263 for fixed worst-width batching. The adaptive-minus-fixed gain is 5.02 percentage points (descriptive 95% calendar-block interval, 2.51–8.60); it is 0.12 points on the original panel (-1.34–1.71). Greedy certifies one more strict external winner than exact adaptation, 208 versus 207. Thus feedback can help model decisions, while exact precision planning has no observed advantage

over greedy for the external tolerance criterion. Online Resource 1, S18 reports the full counts and unchanged tolerance sweep.

When feedback changes the available guarantee. Exact rational checks partition the original 819 blocks into 162 where a fixed batch guarantees half-width, 168 where adaptive querying does so but no deterministic batch does, and 489 where no deterministic two-query policy does. The corresponding external counts are 128, 57 and 94. These are guarantees over the declared uncapped, noiseless simplex, not frequencies of success on the recorded trace.
Among the 165 original and 57 external blocks with matched randomized-batch optimum $1/2$, the greedy first query fails the half-width requirement on 25 and 13 blocks, respectively, even with its optimal conditional second query. Greedy nevertheless succeeds on the recorded trace in 20 of those 25 and 11 of those 13 blocks. The diagnostic identifies a specific benefit of planning the first query: recorded success can conceal an inadequate pre-query guarantee. Across the full panels, this first-query limitation occurs on 28 and 16 adaptive-guaranteed blocks. The supplied interval covers and uncovered-answer witnesses use rational arithmetic on the stored contrast coefficients.

Certifying an insufficient budget. Proposition 2 closes the last branch of Figure 1. All 489 original and 94 external blocks without a deterministic guarantee have twenty-record certificates bounding every randomized adaptive policy's success by $19/20 < 1$. A rational checker reconstructed all 6,588 source coefficient rows and 5,490 first-query covers, then verified 11,660 failing endpoint pairs. The smallest strict normalized failure margins were 0.00031746 and 0.00073903, respectively. These results apply to the uncapped, noiseless model. They justify escalating an insufficient two-reading contract instead of hoping randomization will provide certainty. Online Resource 1 includes every synthetic witness; its default command checks the external panel, and the pinned raw-source route checks both panels.

3.3 Additional information and precision requirements

Absolute precision. The absolute-floor calculation at the primary setting identifies 186 of the 225 baseline reading-saving blocks where feedback remains necessary: all 168 original blocks and 18 external blocks. The other 39 external blocks meet the requirement with a fixed budget of zero (15 blocks), one (14), or two (10) readings. Two further external blocks become two-versus-three separations, giving 20 across the complete external panel. The all-block fixed/adaptive-fallback totals are 2625/2457 original and 378/358 external, or 6.40% and 5.29% fewer readings, respectively. The floor-aware planner certifies the unchanged 57-block external subset with 70 readings, compared with the minimum fixed total of 88 (20.45% fewer). Its exported plans specify all fixed timestamps and adaptive branches; the verifier checks every block's guarantee directly.
Initial loss-scale widths explain the different responses. Among the individually binding contrasts in the 225 blocks, the original median $W_{r,0}$ is 0.12536 squared kWh (interquartile range 0.07424–0.27658); the external median is 0.001905 (0.000741–0.008365). There are 136 and 47 such contrasts, respectively; 69 blocks have no single deletion-binding contrast because of redundant constraints. Online Resource 1 gives every scalar budget, binding contrast and initial width, all seven floor settings, and system-specific acquisition counts.

Threshold stability of the matched certificates. We kept the adaptive branches, equal-weight batch mixtures, and adverse-record distributions fixed and calculated a rational interval $[L, U)$ for each of the 222 matched certificates. Throughout this interval, adaptation guarantees success and the optimal randomized batch has worst-record success exactly $1/2$. The lower endpoint combines the adaptive branch bound with the two-batch covering bound; the upper endpoint is the first threshold at which the saved adverse distribution permits success above $1/2$ for a batch. All 222 intervals contain $1/2$ in their interior. The complete bands $[0.49, 0.51]$, $[0.475, 0.525]$, and $[0.45, 0.55]$ are certified on 179, 119, and 53 blocks, respectively. The symmetric margin around $1/2$ has median 0.0263, interquartile range $[0.0134, 0.0488]$, and minimum 0.00020. These are sufficient ranges for the frozen certificates; outside them this check is inconclusive. Supplement S1 provides the exact endpoints and a solver-free checker that reconstructs all branch and covering inequalities.

Table 8: Blocks retaining adaptive certainty and an exact randomized-batch success optimum of $1/2$ at each precision threshold. The same policies and adverse distributions are used throughout.

Threshold τ	0.45	0.475	0.49	0.50	0.51	0.525	0.55
Original ($n = 165$)	73	120	153	165	148	125	91
External ($n = 57$)	19	30	43	57	56	51	35
Total ($n = 222$)	92	150	196	222	204	176	126

Capacity information versus cohort selection. We recomputed minimum fixed budgets on the same compatible records, retaining the uncapped width target. Table 9 separates excluded members of the original 225 reading-saving blocks from those for which the cap makes a fixed batch sufficient. At $E/2$, only 7 original and 2 external members still require feedback. The full capped counts of 56 and 4 also include 49 and 2 previously unguaranteed blocks that become adaptive-certifiable. At $E/3$, every compatible member of the original 225 becomes fixed-certifiable. The minimum fixed totals on identical compatible subsets fall from 2,509 to 1,840 and from 543 to 424 at $E/2$, and from 2,003 to 768 and from 538 to 220 at $E/3$, for the original and external panels respectively. This attributes the change to both exclusions and genuinely stronger information.

Table 9: Capacity transitions within the original 225 reading-saving blocks. The last two columns partition compatible members. The cap and precision target are the same for both acquisition policies.

Panel	Cap	Reference	Excluded	Fixed ≤ 2	Still needs feedback
Original	$E/2$	168	8	153	7
External	$E/2$	57	17	38	2
Original	$E/3$	168	58	110	0
External	$E/3$	57	17	40	0

Shared-query sensitivity to stored-reading errors. Across all 279 external blocks, the frozen shared policy guarantees that all six normalized comparison widths are at most $1/2$ on 165 blocks under fixed internal-reading errors bounded by 1% of block energy; the best deterministic batch does so on 120. On the 45 additional blocks, feasible fixed-record certificates prove that every randomized batch has worst-case precision success below one. The certified batch success upper bound has minimum and median $1/2$, interquartile range $[1/2, 2/3]$, and maximum $4/5$; 26 bounds are at most $1/2$ and 37 are at most 0.7. These one-sided certificates bound the unrestricted batch optimum. The 45 blocks span all three external systems (11, 11, and 23 blocks).
Capacity information reduces the benefit. With a per-hour cap of half the block energy as well as the same 1% error allowance, a policy frozen from the capacity-aware noiseless planner certifies 213 of the 279 coefficient instances, versus 204 for the best deterministic batch; nine have a certified randomized-batch limitation, with upper bounds between $2/3$ and $7/8$. Only 200 recorded profiles satisfy this illustrative cap. Within that compatible subset the counts are 153 versus 147, with six separations. The original uncapped policy certifies only 191 of the 279 capped instances, which motivates using declared capacity information when planning. We evaluate adaptive policies frozen from the respective noiseless planners; the fixed-batch comparator is optimized for the error model. All policies are fixed before the fine readings are opened.
The comparison uses two shared readings, uncapped noiseless initial widths for normalization, exact block endpoints, and the same permitted queries for every method. A separately implemented rational checker verifies the saved contrast coefficients, branch coverage, every reported precision status, and the complete fixed-error adverse records. Supplementary methods (Online Resource 1, S8) give the certificate construction.

Reproducibility and numerical checks. The supplement connects version-pinned source records, frozen forecasts, contrast coefficients, saved policies and reported outcomes. A fresh both-source run reproduced all 1,098 outcome rows exactly, including the 165 original and 57 external matched-half certificates. The original source route rejects mismatched files; its raw records are regenerated locally. Default offline replay uses the bundled inputs and is separate from raw regeneration and fresh model fitting.
A separately written checker reconstructs the policies and summaries. On 18 fixed representative blocks, 19,440 endpoint LPs and 6,480 conditional LPs have maximum discrepancies of 2.22×10^{-16} and 1.11×10^{-16}, against a 10^{-8} check tolerance. Exact rational covering and adverse-record inequalities establish the matched precision claims. An optimal-face diagnostic also checks that arbitrary selection among mean-optimal batch mixtures does not explain their lower recorded success. Online Resource 1, S8 supplies the full checks and the results-to-command map identifies each reproduction entry point.

3.4 What the retrieval interface changes

The HeatpumpMonitor API permits full-range retrieval, so we tested it directly (OpenEnergyMonitor, 2026b,c). Two repetitions of five policies on one preselected task from each external system produced 30 trials and 60 HTTP requests, including common endpoint reads. Bulk retrieval uses one request and achieves zero terminal width; adaptation uses two requests.

Mean measured incremental I/O time, excluding client throttle waiting, is 0.331 seconds for bulk and 0.497 seconds for adaptation. Mean response bodies are 226.7 and 104.3 bytes, respectively. We recommend bulk when the full response is admissible. The selective planner instead computes the readings required under a declared access contract. Full measurements and cached responses appear in Online Resource 1, S15.

4 Discussion

4.1 Planning the readings needed for assurance

The auditor supplies the comparisons, available information and permissible measurements. Our method returns the minimum sufficient fixed reading budget and checks whether feedback reduces it. The reading budget is an output of this calculation, so the auditor need not first impose a two-reading quota. Each saving exchanges one scalar reading for an additional information round. With declared charges $c_{\text{read}} > 0$ per reserved reading and $c_{\text{round}} \geq 0$ per round, the two assured plans cost $3c_{\text{read}} + c_{\text{round}}$ and $2c_{\text{read}} + 2c_{\text{round}}$. Feedback is preferable exactly when $c_{\text{round}} < c_{\text{read}}$; ties favor the one-round fixed plan. This is an analytic choice between these plans under the same exact information model. An admissible bulk action is compared separately.

Theorem 2 explains why sharing readings matters. Two simultaneous comparisons can remove the scalar two-query success floor: adaptive certainty coexists with optimal batch success $1/m$, although each contrast separately admits a sufficient fixed batch. The adverse records force all m batches even for positive worst-record success and identify the unique optimal mixture. The output dimension and query budget stay fixed as the dictionary grows; three fixed readings suffice in this construction.

The scalar-budget checks connect this construction to real forecast coefficients: most reading-saving blocks are already certifiable pair by pair, but require feedback when the readings must serve every pair. The exact threshold intervals establish where this conclusion remains stable. Recorded gains are larger over fixed batching than over greedy adaptation; they describe performance on these panels rather than the worst-case separation proved by the certificates.

Absolute precision changes which readings are worth acquiring. At the declared training-scaled floor, all 57 external cases retain assurance with at most two readings, but 39 can use a simpler fixed plan. The floor-aware planner therefore reserves 70 readings rather than the minimum fixed total of 88 within this unchanged cohort. Across the complete external panel the reduction is 5.29%. The half-width fraction and absolute floor are reporting targets. Relating them to a financial materiality threshold requires application-specific calibration.

4.2 Scope and protocol choice

The main planner uses two cumulative-prefix queries and an exact nonnegative total. Its reading budget measures disclosed scalar values; the API control measures requests and I/O time. An unrestricted range request returns the complete record faster on the tested interface, so bulk is preferable when admissible. Online Resource 1, S16 evaluates a separate stopping rule for tolerance-based model choice. With endpoints already retained, it certifies an acceptable model before fetching finer data on 8/819 original and 198/279 external blocks. Its system-level

loss-scale table explains the tolerance; cold-start and whole-period comparisons account for endpoint setup and range batching. This is a supplementary acquisition option, distinct from joint-precision planning.

Additional information can change the plan. Of the 225 baseline saving blocks, 75 are excluded by the stronger $E/3$ cap and all 150 compatible blocks become fixed-certifiable. Thus capacity removes the feedback requirement on those same records, independently of the exclusion effect.

Bounded errors can increase the sufficient reading budget. Table 1 states these sensitivity models, including exact retained endpoints and uncalibrated error allowances. The source panels each contain three systems and were reused retrospectively. The descriptive intervals summarize these panels; dependence beyond the resampled calendar blocks remains unmodeled. Population generalization requires additional systems.

Precision limits how much an audit's reported loss difference can change when the missing readings arrive. A model decision also depends on the intervals' positions. At the training-scaled tolerance, adaptive and greedy querying each certify an acceptable model on 277 external blocks, versus 263 for fixed batching. The strict-winner counts favor greedy by one block. These outcomes distinguish the benefit of feedback from the value of exact planning for the precision requirement. The tolerance and charges remain choices for an audit service; the study measures data requirements and API behavior, not a deployed financial return. Online Resource 1 contains the broader scalar theory, coordinate-recovery solver and protocol-selection cost sensitivity. A next step is to develop scalable planners for larger adaptive reading budgets.

5 Conclusions

Sharing a reading budget can make feedback necessary for an assurance requirement that each comparison can meet separately. With two affine targets and two coordinate queries, adaptive success can be one while optimal randomized-batch success is $1/m$. The same construction covers all pairs among three models.

Our exact planner and verifiable certificates turn this boundary into a reading-budget decision. The meter study establishes where two adaptive readings replace the minimum three fixed readings and isolates the shared-budget mechanism through separate scalar checks.

Absolute-scale and capacity analyses identify where fixed readings suffice instead. The resulting plans specify the readings and feedback required for a stated assurance level, alongside the option of an admissible bulk export.

Supplementary Materials

Online Resource 1 is supplied as Supplemental Data S1 and S2. Extract both ZIP files into the same directory using the Python extraction commands in the accompanying README. Together they contain the editable manuscript, executable methods, frozen forecasts, source identifiers, case ledgers, exact witnesses and numerical checks. Start with `README.txt`; `CURRENT_RESULTS_MAP.txt` maps each current table and verification claim to its command, inputs and outputs. The default verification replays saved forecasts. Separate commands cover raw-input regeneration and fresh fitting. The indexed extended methods retain full proofs, solver

scaling, forecasting protocols and historical controls.

Additional Information and Declarations

Data and Code Availability

The numerical code and detailed outputs are supplied in Online Resource 1. The central study uses BLEM and HeatpumpMonitor. The package bundles the attributed HeatpumpMonitor records and frozen forecasts. BLEM counter traces are regenerated from the pinned public QuantLet repository (`https://github.com/QuantLet/BLEM`), accompanying Kostmann and Härdle (2019; `https://doi.org/10.3390/en12142718`). The regeneration commands reject inputs with mismatched hashes. These raw traces are not redistributed because redistribution permission has not been established. Bundled Portugal and household analyses are historical supporting controls, separate from the central two-panel study. The complete polytopes, rational certificates, numerical tolerances and paired decision outcomes are supplied in Supplemental Data S1 and S2; `CURRENT_RESULTS_MAP.txt` identifies the reproduction route for each result.

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Competing Interests

The author declares no competing interests.

Author Contributions

Jishan Kapoor conceived and designed the research, developed the methodology, conducted the computational analyses, interpreted and verified the results, prepared and revised the manuscript, and approved the final version.

Research Transparency

This work uses public data and constructed mathematical instances, with no newly recruited participants. The shared-query analyses are retrospective on both source panels. Separate checker implementations provide within-study verification; they are not external independent replication.

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